

Bit Manipulation – 1

Bitwise Operators

A	B	A & B	A B	A ^ B	~B
0	0	0	0	0	1
1	0	0	1	1	1
0	1	0	1	1	0
1	1	1	1	0	0

Properties:

Q1. What is $A|0 = A$

```
A = 1010
    0000
    1010
```

Q2. $A|A = A$

Q3. $A^0 = A$

$1^0 = 1$ and $0^0 = 0$. Opposite value gives 1. Same value gives 0.

Q4. $A^A = 0$

$1^1 = 0$ and $0^0 = 0$

Q5. $A \& 0 = 0$

Q6. $A \& A = A$

$0 \& 0 = 0$

$1 \& 1 = 1$

7. Last bit determines even or odd.

If last bit zero $\rightarrow$ the number is even. If last bit is one $\rightarrow$ the number is odd

8. How to check last bit?

$A \& 1 \rightarrow 0 \rightarrow$ even no.

$A \& 1 \rightarrow 1 \rightarrow$ odd no.

If you bit & with 1 $\rightarrow$ everything will become 0 except the last bit.

```
A=5
 101
& 001
 001
```

```
A = 20
 10100
& 00001
 00000
```

9. $A\%2 == 0 \rightarrow$ takes some extra CPU cycles but $A\&1$ bitwise operation is faster than arithmetic operation but TC remains same which is $O(1)$.

10. Commutative Property

```
A & B = B & A
A | B = B | A
A ^ B = B ^ A
```

11. Associative Property

$$(A \& B) \& C = A \& (B \& C)$$

$$(A | B) | C = A | (B | C)$$

$$(A \wedge B) \wedge C = A \wedge (B \wedge C)$$

Quiz: What would be the output of $a \wedge b \wedge a \wedge d \wedge b$?

$$(a \wedge a) \wedge (b \wedge b) \wedge d$$

$$0 \wedge 0 \wedge d$$

$$0 \wedge d$$

$$d$$

1. Problem Statement

Given an integer array where every no. occurs twice except one number. Find that unique no.?

Pseudocode

```
ans = A[0]
for (i = 1 to N-1) {
  ans ^= A[i]
}
return ans
```

$$TC = O(N)$$

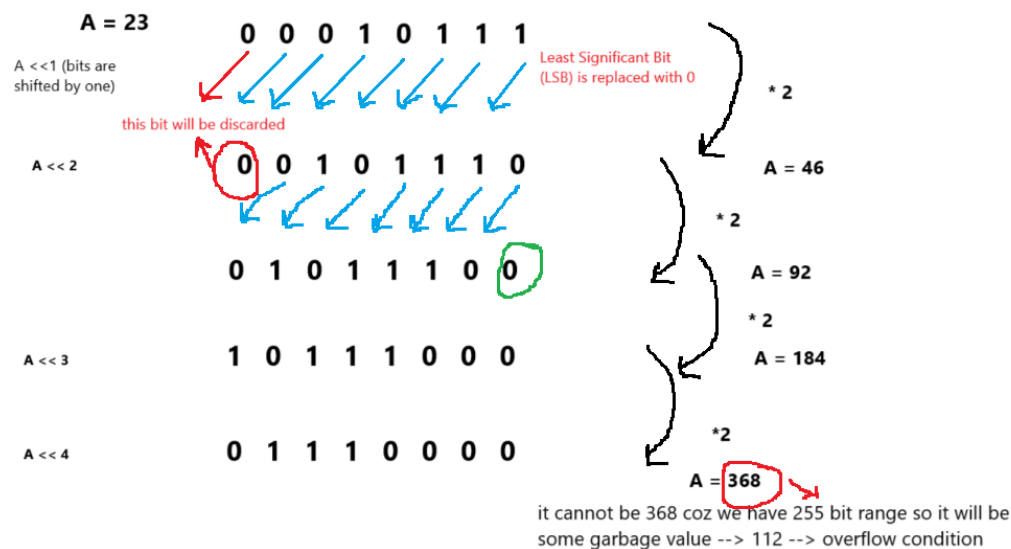
$$SC = O(1)$$

12. Left Shift Operator (<<)

It shifts the numbers in the left by one. Lets assume our operators are 8 bits.

8 bits can be from 0 to 255 or -128 to 127. Assuming unsigned bits 0 to 255.

A = 23.



If I left shift one time $N \ll 1 = N * 2$

If I left shift K times $N \ll k = N * 2^k$

$$1 \ll n = 2^n$$

13. Check for overflow

$x * 2 > \text{Integer max}$

$$x > \frac{\text{Integer Max}}{2}$$

$$x < \frac{\text{Integer Min}}{2}$$

14. Right Shift Operator (>>)

Right Shift Operator shifts the numbers in the right by one.

A = 20	0 0 0 1 0 1 0 0		
A >> 1	0 0 0 0 1 0 1 0	20/2 = 10	
A >> 2	0 0 0 0 0 1 0 1	10/2 = 5	
A >> 3	0 0 0 0 0 0 1 0	5/2 = 2	
A >> 4	0 0 0 0 0 0 0 1	2/2 = 1	
A >> 5	0 0 0 0 0 0 0 0	1/2 = 0	

$$N \gg 1 = N/2$$

$$N \gg k = N/2^k$$

This is no overflow condition

QUIZ: What will be the output of $1 \ll 3$

$$1 \ll n = 2^n$$

when $n = 3$,

$$2^3 = 8$$

15. Power of left shift operator

1. OR

$N | (1 \ll k) \rightarrow$ set Kth bit of N

2. XOR

$N \wedge (1 \ll k) \rightarrow$ toggle kth bit

3. AND



16. Unset a kth bit

```
def unset(N, K){
    X = N & (1 << k)           //unset if X ==0, kth bit is already unset
    if (X > 0) {
        N = N^(1<<k)
    }
}
```

17. Check if kth bit is set or unset?

```
def isset(N, K){
    X = N & (1 << k)
    if(X ==0) return False
    return True
}
```

2. Problem Statement

Given an integer N, count total set bits in N (consider N as 32 bit integer)

Brute Force - Pseudocode

If 32 bit integer $\rightarrow$ 0 to 31 range

```
ans = 0
for (i = 0 to 31){
    if(N & (1 << i) >0){
        ans++
    }
}
return ans;
```

TC = $O(1)$ $\rightarrow$ coz we are running the loop 31 times

SC = $O(1)$

This method is not that optimized as below method. If $N = 1$, this loop will run 31 times. If $N = 4$, this loop will run 31 times.

Optimized Code – Pseudocode

```
def countbits()
ans = 0
while(N > 0) {
    if(N & 1){
        ans ++
    }
    N = N >>1
}
return ans;
```

Explanation:

$N \& 1$ $\rightarrow$ if last bit is one, it will give 1 $\rightarrow$ this will check whether a bit is set or unset.

$N = N \gg 1$ $\rightarrow$ in each iteration we are dividing $N = N/2$ $\rightarrow$ this will drop MSB (most significant bit) each time until $N > 0$.

If $N = 1$, this loop will run 1 time,

if $N=4$, this loop will run 3 time only instead of 31 every time as in previous code.

TC = $O(\log N)$

SC = $O(1)$

3. Problem Statement

For each train, there is a special no. Special no.: 28 bit no. where ith bit is set then train runs on that day. Write a code which will tell which train runs the most.

$A = [20, 2, 10]$

20 $\rightarrow$ 1 0 1 0 0 $\rightarrow$ This train runs on 2 days

7 $\rightarrow$ 1 1 1 $\rightarrow$ This train runs on 3 days

10 $\rightarrow$ 1 0 1 0

Count the set bits of all trains & return the train having most set bit count.

def findTrain(a[], N){